

18. Design a fully connected network using RTL

AIM

To design and simulate a Fully Connected (FC) neural network layer with five input neurons and one output neuron using RTL (Verilog HDL).

SOFTWARE REQUIRED

- Xilinx Vivado Design Suite
- Vivado Simulator (XSIM)
- Verilog HDL
- Windows / Linux Operating System

PROCEDURE

1. Open Xilinx Vivado and create a new RTL project.
2. Select the target FPGA device or board.
3. Create a Verilog module for the Fully Connected layer with five input neurons.
4. Assign individual weights to each input neuron and a single bias to the output neuron.
5. Implement the multiply–accumulate (MAC) operation using fixed-point arithmetic.
6. Add the bias value to the accumulated sum to generate the final output.
7. Write a testbench to apply different signed input and weight values.
8. Simulate the design using Vivado Simulator.
9. Observe the output waveform and verify the computed output with manual calculations.

Verilog program for fully connected layer

```
module fully_connected_5in (  
    input signed [7:0] x0, x1, x2, x3, x4, // 5 inputs  
    input signed [7:0] w0, w1, w2, w3, w4, // 5 weights  
    input signed [15:0] bias, // single bias  
    output signed [17:0] y // output  
);  
  
wire signed [15:0] m0, m1, m2, m3, m4;  
  
wire signed [17:0] sum;  
  
assign m0 = x0 * w0;  
assign m1 = x1 * w1;
```

```

assign m2 = x2 * w2;
assign m3 = x3 * w3;
assign m4 = x4 * w4;
assign sum = m0 + m1 + m2 + m3 + m4;
assign y = sum + bias;
endmodule

```

Testbench program for fully connected layer

```

`timescale 1ns / 1ps

module tb_fully_connected_5in;

    reg signed [7:0] x0, x1, x2, x3, x4;
    reg signed [7:0] w0, w1, w2, w3, w4;
    reg signed [15:0] bias;
    wire signed [17:0] y;

    // Instantiate the DUT
    fully_connected_5in uut (
        .x0(x0), .x1(x1), .x2(x2), .x3(x3), .x4(x4),
        .w0(w0), .w1(w1), .w2(w2), .w3(w3), .w4(w4),
        .bias(bias),
        .y(y)
    );

    initial begin
        // Test Case 1
        x0 = 8'd10; x1 = 8'd20; x2 = 8'd30; x3 = 8'd40; x4 = 8'd50;
        w0 = 8'd1; w1 = 8'd1; w2 = 8'd1; w3 = 8'd1; w4 = 8'd1;
        bias = 16'd10;
        #10;

        // Test Case 2
        x0 = 8'd5; x1 = 8'd10; x2 = 8'd15; x3 = 8'd20; x4 = 8'd25;
        w0 = 8'd2; w1 = 8'd2; w2 = 8'd2; w3 = 8'd2; w4 = 8'd2;
        bias = 16'd0;
        #10;
    end
endmodule

```

```

// Test Case 3 (signed values)

x0 = -8'd10; x1 = 8'd20; x2 = -8'd5; x3 = 8'd15; x4 = 8'd10;

w0 = 8'd1; w1 = 8'd1; w2 = 8'd1; w3 = 8'd1; w4 = 8'd1;

bias = 16'd5;

#10;

$stop;

end

endmodule

```

Output

Name	Value	0 ns	20 ns
>  x0[7:0]	246	10	5
>  x1[7:0]	20	20	10
>  x2[7:0]	251	30	15
>  x3[7:0]	15	40	20
>  x4[7:0]	10	50	25
>  w0[7:0]	1	1	2
>  w1[7:0]	1	1	2
>  w2[7:0]	1	1	2
>  w3[7:0]	1	1	2
>  w4[7:0]	1	1	2
>  bias[15:0]	5	10	0
>  y[17:0]	35	160	150

RESULT

The Fully Connected Network with five input neurons was successfully designed and simulated using RTL (Verilog HDL). The circuit correctly performed multiplication of each input with its corresponding weight, accumulated the results, and added the bias to generate the final output. The simulation results matched the expected theoretical values, confirming the correct functionality of the Fully Connected layer.